

Week 3 lecture 1

Programmatic Problem solving

What is a problem?

- ◆ Some well defined equation?
- ◆ Building a deck?
- ◆ An inquiry starting from given conditions to investigate or demonstrate a fact, result, or law.

Our problems

Some well defined enquiry of fabrication or
virtualization

How can we solve problems?

(Part 1)

1. Assess the problem
2. Define the problem
3. Gather Data

How can we solve problems?

(Part 2)

1. Develop Algorithm
2. Program Algorithm
3. Evaluate
4. Implantation of program

Assess the problem

- ◆ Trying to define the problem space
- ◆ Big picture
- ◆ All variables

Define the problem

- ◆ Divide the problem space up
- ◆ subsets of all variables

Gather Data

- ◆ Further analyze the problem and variables
- ◆ Choose a subset of variables & investigate

Develop Program

- ◆ Pre-determined logic
- ◆ Pseudocode

Program Algorithm

- ◆ Pick a programming language
- ◆ Transform you pseudocode
 - ◆ Pseudocode → programming language

Evaluate & implant

- ◆ Testing
- ◆ deployment
- ◆ Out of scope

"The difference between theory and practice is
smaller in theory than it is in practice."

—Albert Einstein

Functions & Files

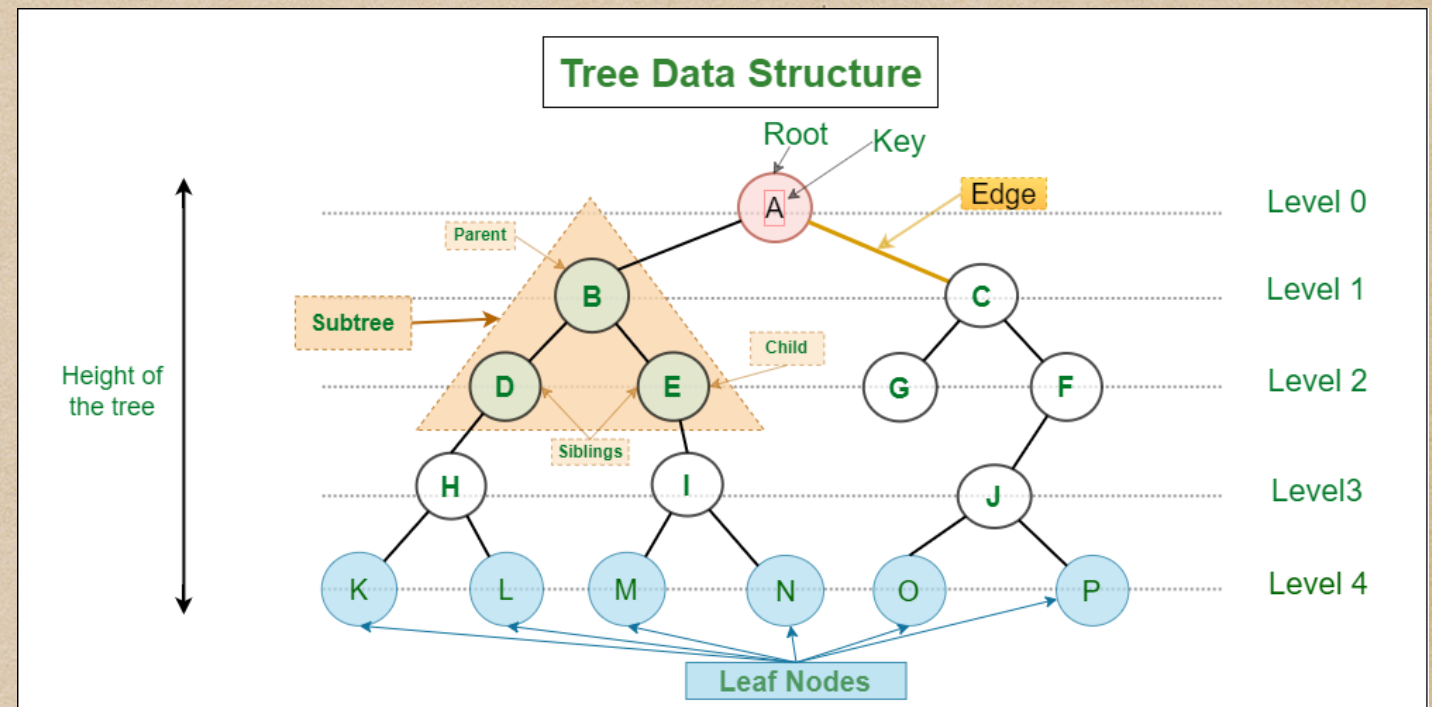
- ◆ What is a file?
- ◆ Functions and how to use them

File?

Finite, small, data blocks, that are easily
searchable.

File system

- ◆ Tree structure
- ◆ Node = file
- ◆ Branch = encapsulation



File structure

- ◆ Name
- ◆ Location
- ◆ Extension (type)

“Engineers like to solve problems. If there are no problems handily available, they will create their own problems.”

- Scott Adams

Functions

What is a function

Portable, invokable, sequence of logic

Function

- ◆ Invokable
- ◆ Portable
- ◆ Sequence of logic

Function structure

- ◆ Name
- ◆ Input collection
- ◆ Single Output
- ◆ Logic

Questions